F18CE Week 4

Lagrange Multiplier

Lagrange multiplier with one constraint (Theorem 5.7)

Let $f(\mathbf{x})$ be the objective function subject to one constraint $g(\mathbf{x}) = 0$.

The goal is to find an optimum solution/point $\mathbf{x} \in \mathbb{R}^n$ that satisfies the above.

Suppose $\mathbf{x} \in \mathbb{R}^n$ exists. To determine $\mathbf{x}$:

- Find all values of $\mathbf{x}$ such that $\nabla(f(\mathbf{x}) + \lambda g(\mathbf{x})) = 0$ and $g(\mathbf{x}) = 0$.
- The largest of these values is $f_{\max}$ and the smallest is $f_{\min}$.

Remark: 1. It is clear that one can use $\nabla(f(\mathbf{x}) - \lambda g(\mathbf{x})) = 0$.

2. Further investigation need to be done if we only obtain one solution.

Example 1 (Example 5.8)

Use the method of the Lagrange multiplier to minimise

$f(x,y,z) = x^2 + y^2 + z^2$ subject to $2x + 3y - z = 5$.

Solution. Let $g(x,y,z) = 2x + 3y - z - 5$ so that $g(x,y,z) = 0$.

Step 1. $\nabla(f + \lambda g) = 0 \Rightarrow \begin{cases} 2x + 2\lambda = 0 & (1) \\ 2y + 3\lambda = 0 & (2) \\ 2z - \lambda = 0 & (3) \end{cases}$ and $2x + 3y - z = 5$ (4).

Step 2. Substitute $\begin{cases} x = -\lambda \\ y = -\frac{3}{2}\lambda \\ z = \frac{1}{2}\lambda \end{cases}$ into (4):

$$-2\lambda + 3(-\tfrac{3}{2}\lambda) - \tfrac{1}{2}\lambda = 5 \Rightarrow \lambda = -\tfrac{5}{7}\,.$$

So, $(x, y, z) = (\tfrac{5}{7}, \tfrac{15}{14}, -\tfrac{5}{14})\,.$

Step 3. Check that $f(\frac{5}{7}, \frac{15}{14}, -\frac{5}{14}) = (\frac{5}{7})^2 + (\frac{15}{14})^2 + (-\frac{5}{14})^2 = \frac{25}{14}$

This is a minimum of f subject to $2x + 3y - z = 5$.

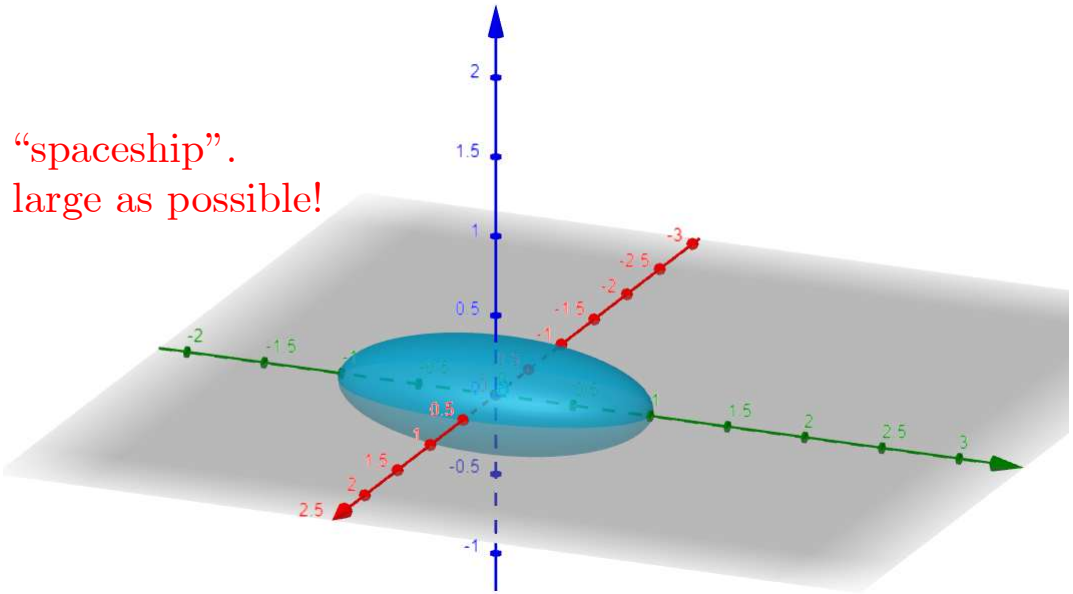
To verify this, we choose $(x, y, z) = (0, 0, -5)$ (satisfies $g(x, y, z) = 0$)

and check that $f(0, 0, -5) = 25 > \frac{25}{14}$.

Example 2 (Example 5.9)

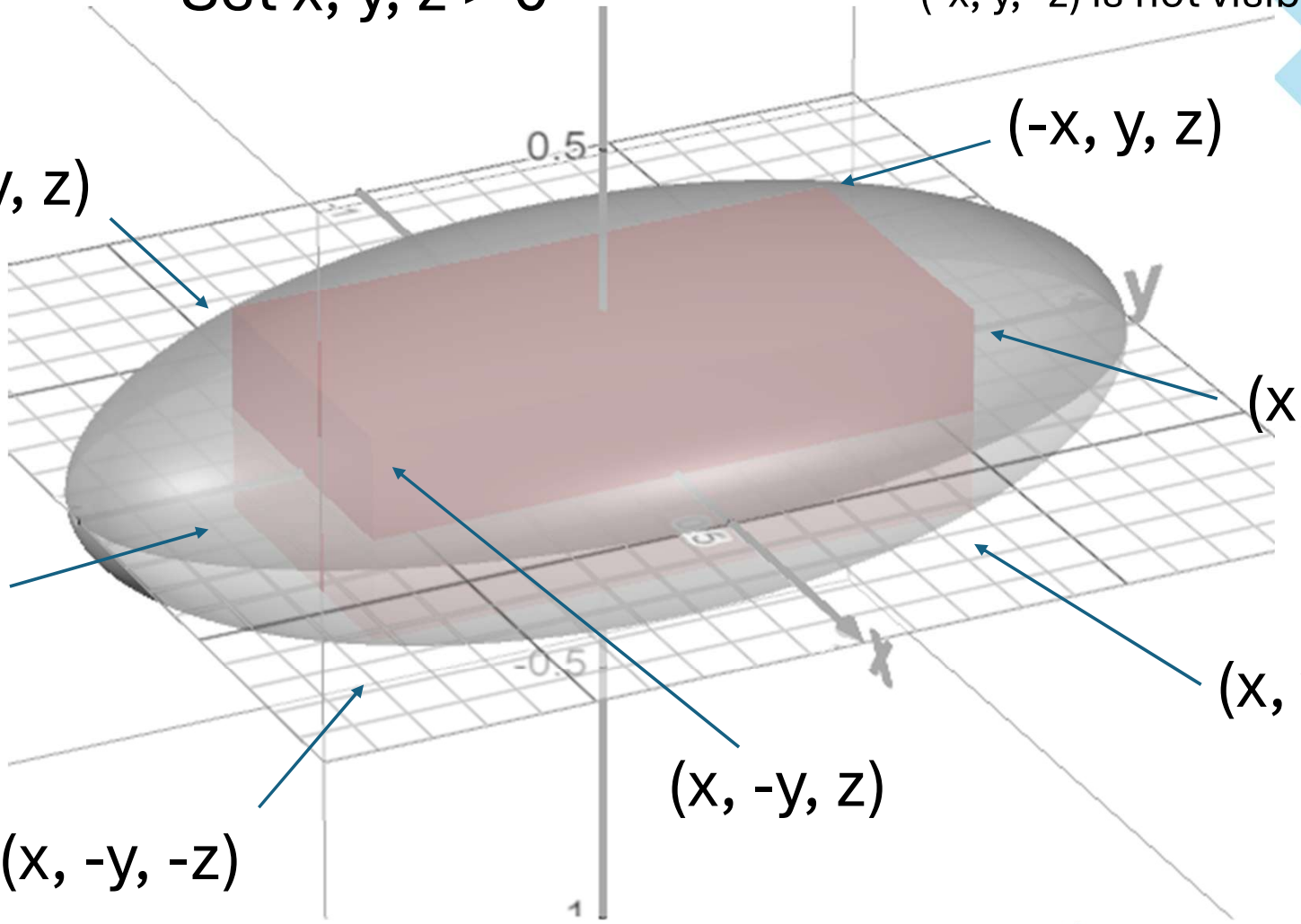
Find the rectangular box of maximal volume with sides parallel to the coordinate planes which can be contained inside the ellipsoid described via the equation $4x^2 + y^2 + 9z^2 = 1$.

Put a box inside this "spaceship".
The box has to be as large as possible!



Set x, y, z > 0

(-x, y, -z) is not visible here



(-x, y, z)

(-x, -y, z)

(x, y, z)

(-x, -y, -z)

(x, y, -z)

(x, -y, z)

(x, -y, -z)

Solution. Let $x, y, z > 0$. Maximise $V = 8xyz$ subject to

$$g(x, y, z) = 4x^2 + y^2 + 9z^2 - 1 = 0.$$

Step 1. $\nabla(V + \lambda g) = 0 \Rightarrow \begin{cases} 8yz + 8x\lambda = 0 \\ 8xz + 2y\lambda = 0 \\ 8xy + 18z\lambda = 0 \end{cases} \Rightarrow \begin{cases} 4xyz = -4x^2\lambda & (1) \\ 4xyz = -y^2\lambda & (2) \\ 4xyz = -9z^2\lambda & (3) \end{cases}$

and $g(x, y, z) = 0$ (4).

Step 2. $(1)+(2)+(3): 12xyz = -\lambda(4x^2+y^2+9z^2) = -\lambda \times 1 = -\lambda$.

If $\lambda = 0$, then $xy = xz = yz = 0$. This is impossible since $x, y, z > 0$.

So $\lambda \neq 0$ and thus

$$\begin{cases} 4xyz = -\frac{1}{3}\lambda = -4x^2\lambda \\ 4xyz = -\frac{1}{3}\lambda = -y^2\lambda \\ 4xyz = -\frac{1}{3}\lambda = -9z^2\lambda \end{cases} \Rightarrow \begin{cases} x^2 = \frac{1}{12} \\ y^2 = \frac{1}{3} \\ z^2 = \frac{1}{27} \end{cases}$$

$\Rightarrow (x, y, z) = \left(\frac{1}{2\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{3\sqrt{3}}\right)$ since we set $x, y, z > 0$.

Step 3. We see that $V_{\max} = V\left(\frac{1}{2\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{3\sqrt{3}}\right) = \frac{4}{9\sqrt{3}}$.

One can verify using an appropriate point (a, b, c) to show that $V(a, b, c) < V_{\max}$.

Example 3 (Example 5.10)

Find the maximum and minimum of $f(x,y) = x^2 - xy + y^2 - 3x$ subject to $x^2 + y^2 - 9 \leq 0$.

Solution. Let $g(x,y) = x^2 + y^2 - 9$.

Case 1. Suppose we optimise $f(x,y) = x^2 - xy + y^2 - 3x$ subject to $g(x,y) = 0$. Then we have

$$\begin{cases} 2x - y - 3 + 2x\lambda = 0 \\ -x + 2y + 2y\lambda = 0 \end{cases} \Rightarrow \begin{cases} 2x(1+\lambda) = y + 3 \\ 2y(1+\lambda) = x \end{cases} \Rightarrow \begin{cases} 2xy(1+\lambda) = y^2 + 3y \\ 2xy(1+\lambda) = x^2 \end{cases}$$

So, we have $x^2 = y^2 + 3y$. Substitute this into $g(x,y) = 0$ to obtain

$$2y^2 + 3y - 9 = 0 \Rightarrow y = \tfrac{3}{2}, -3.$$

When $y = \frac{3}{2}$, $x^2 = \frac{9}{4} + \frac{9}{2} = \frac{27}{4} \Rightarrow x = \pm\frac{3\sqrt{3}}{2}$.

So, $(x, y) = \left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right), \left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$.

When $y = -3$, $x^2 = 0 \Rightarrow x = 0$.

So, $(x, y) = (0, -3)$.

We compare:

$$f\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right) = 9 - \frac{27\sqrt{3}}{2} < f(0, -3) = 9 < f\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right) = 9 + \frac{27\sqrt{3}}{2}.$$

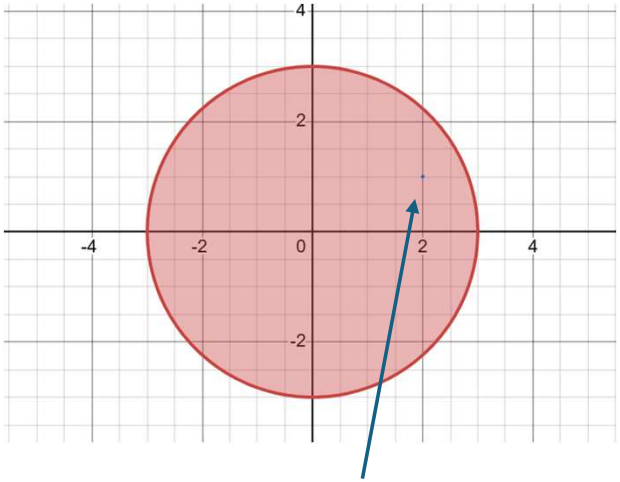
Case 2. Suppose we optimise $f(x,y) = x^2 - xy + y^2 - 3x$ without any constraint. Note that

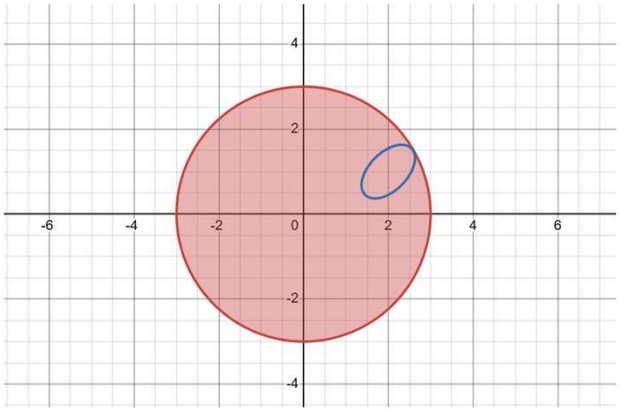
$$\begin{aligned} f(x,y) &= x^2 - xy + y^2 - 3x \\ &= \left(\tfrac{1}{2}x - y\right)^2 + \tfrac{3}{4}x^2 - 3x \\ &= \left(\tfrac{1}{2}x - y\right)^2 + \tfrac{3}{4}(x-2)^2 - 3 \geq -3 \end{aligned}$$

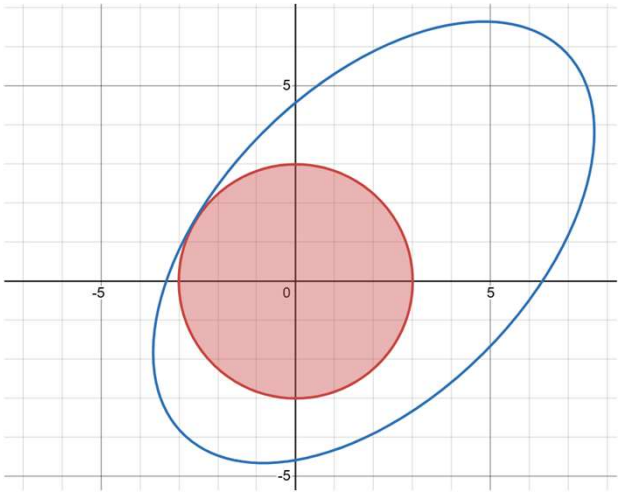
Then $f = -3$ when $(x,y) = (2,1)$. Check that $g(2,1) = -4 < 0$, which mean $(2,\ 1)$ satisfies $g(x,y) = x^2 + y^2 - 9 \leq 0$. Now compare

$$f(2,1) = -3 < f\left(\tfrac{3\sqrt{3}}{2}, \tfrac{3}{2}\right) = 9 - \tfrac{27\sqrt{3}}{2} < f(0,-3) = 9 < f\left(-\tfrac{3\sqrt{3}}{2}, \tfrac{3}{2}\right) = 9 + \tfrac{27\sqrt{3}}{2}.$$

Hence, $f_{\min} = -3$ at $(2, 1)$, and $f_{\max} = 9 + \frac{27\sqrt{3}}{2}$ at $\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$.







$f\left(2,1\right) = -3$ $\quad$ $f\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right) = 9 - \frac{27\sqrt{3}}{4} = -2.69$ $\quad$ $f\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right) = 9 + \frac{27\sqrt{3}}{4} = 20.69$

Lagrange multiplier with more than one equation constraint

Let $\mathbf{x} \in \mathbb{R}^n$. To find the maximum and minimum value of $f(\mathbf{x})$ subject to the constraints $g_i(\mathbf{x}), i = 1,...,m$ (assume that these extreme values exist and $\nabla g_i(\mathbf{x}) \neq \mathbf{0}$): Then we find all values of $\mathbf{x}$ and λ_i such that

$$\nabla f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\mathbf{x}) = \mathbf{0} \;\&\; g_i(\mathbf{x}) = 0 \;\forall i = 1,...,m\,.$$

Evaluate f at all the points $\mathbf{x}$ that result from the above. The largest of these values is the maximum value of f ; the smallest is the minimum value of f.

Example 4

Maximise $f(x,y,z) = x + 2y + 3z$ subject to

$$g_1(x,y,z) = x - y + z - 1 = 0 \text{ and}$$

$$g_2(x,y,z) = x^2 + y^2 - 1 = 0$$

Solution.

$$\nabla\left(f + \lambda_1 g_1 + \lambda_2 g_2\right) = 0 \Rightarrow \begin{pmatrix} 1 + \lambda_1 + 2\lambda_2 x \\ 2 - \lambda_1 + 2\lambda_2 y \\ 3 + \lambda_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

So we have $\lambda_1 = -3$ and therefore

$$\begin{cases} -2 + 2\lambda_2 x = 0 \\ 5 + 2\lambda_2 y = 0 \end{cases} \Rightarrow \begin{cases} \lambda_2 x = 1 \\ \lambda_2 y = -\frac{5}{2} \end{cases} \Rightarrow \begin{cases} \lambda_2 xy = y \\ \lambda_2 xy = -\frac{5}{2}x \end{cases} \Rightarrow -5x = 2y\,.$$

Sub into $x^2 + y^2 = 1$ to get $\frac{4}{25}y^2 + y^2 = 1 \Rightarrow y^2 = \frac{25}{29} \Rightarrow y = \pm\frac{5}{\sqrt{29}}$.

So, $(x, y, z) = \left(-\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}}, 1 + \frac{7}{\sqrt{29}}\right), \left(\frac{2}{\sqrt{29}}, -\frac{5}{\sqrt{29}}, 1 - \frac{7}{\sqrt{29}}\right)$ from $x - y + z = 1$.

$$f\left(-\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}}, 1 + \frac{7}{\sqrt{29}}\right) = 3 + \sqrt{29} = \max$$
$$f\left(\frac{2}{\sqrt{29}}, -\frac{5}{\sqrt{29}}, 1 - \frac{7}{\sqrt{29}}\right) = 3 - \sqrt{29} = \min$$

Alternative: Maximise $x + 2y + 3(1 - x + y) = -2x + 5y + 3$

subject to $x^2 + y^2 = 1$.

$$\begin{cases} -2 + 2x\lambda = 0 \\ 5 + 2y\lambda = 0 \end{cases} \Rightarrow \begin{cases} 2xy\lambda = 2y \\ 2xy\lambda = -5x \end{cases} \Rightarrow 5x + 2y = 0$$

Example 5

Optimise $f(x,y,z) = x^2 + y^2 + z^2$ subject to the constraints

$$0 = g(x,y,z) = x^2 + y^2 - z^2 \quad (1)$$
$$0 = h(x,y,z) = x - 2z - 3 \quad (2)$$

Solution. $\nabla\Big(f + \lambda g + \mu h\Big) = 0 \Rightarrow \begin{pmatrix} 2(1+\lambda)x + \mu \\ 2(1+\lambda)y \\ 2(1-\lambda)z - 2\mu \end{pmatrix} = 0$

$$\begin{cases} y = 0 \text{ or } \lambda = -1 & (3) \\ \mu = -2(1+\lambda)x = (1-\lambda)z & (4) \end{cases}$$

Note that $\lambda = -1$ and $y = 0$ contradicts to this statement.

From (3)

If $\lambda = -1 \underset{(4)}{\Rightarrow} \mu = 0 \underset{(4)}{\Rightarrow} z = 0 \underset{(2)}{\Rightarrow} x = 3 \underset{(1)}{\Rightarrow} y = \pm i\sqrt{3}$ (this is impossible).

If $y = 0 \underset{(1)}{\Rightarrow} x = \pm z \underset{(2)}{\Rightarrow} \pm z - 2z = 3 \Rightarrow (x, z) = (-3, -3), (1, -1)$.

Since $f(1, 0, -1) = 2 < 18 = f(-3, 0, -3)$, it follows that

$f_{\min} = 2$ when $(1, 0, -1)$

$f_{\max} = 18$ when $(-3, 0, -3)$

Example 6 (Alternative solution for Example 5)

Optimise $f(x,y,z) = x^2 + y^2 + z^2$ subject to the constraints

$$0 = g(x,y,z) = x^2 + y^2 - z^2 \quad (1)$$

$$0 = h(x,y,z) = x - 2z - 3 \quad (2)$$

Alternative. Combine the constraints to obtain

$$x^2 + y^2 = \tfrac{1}{4}(x-3)^2 \Rightarrow l(x,y) = 3x^2 + 4y^2 + 6x - 9 = 0 \quad (3).$$

Optimise $F(x,y) = x^2 + y^2 + \tfrac{1}{4}(x-3)^2 = \tfrac{5}{4}x^2 + y^2 - \tfrac{3}{2}x + \tfrac{9}{4}$ subject to (3).

$$\nabla\Big(F + \gamma l\Big) = 0 \Rightarrow \begin{pmatrix} \frac{5}{2}x - \frac{3}{2} + 6\gamma(x+1) \\ 2y + 8\gamma y \end{pmatrix} = 0$$

$$\begin{cases} y = 0 \text{ or } \gamma = -\frac{1}{4} & (4) \\ 5x - 3 + 12\gamma(x+1) = 0 & (5) \end{cases}$$

From (4), $y = 0 \underset{(3)}{\Rightarrow} x^2 + 2x - 3 = 0 \Rightarrow x = 1, -3$.

From (4),

$\gamma = -\frac{1}{4} \underset{(5)}{\Rightarrow} 5x - 3 = 3(x+1) \Rightarrow x = 3 \underset{(3)}{\Rightarrow} 27 + 4y^2 + 12 - 9 = 0 \Rightarrow 4y^2 = -30$,

this is impossible.

Hence, $F(1,0) = 2 < F(-3,0) = 18$.

Example 7

Find the distance from the origin to the curve that is the intersection of the two surfaces

$$z^2 = x^2 + y^2 \text{ and } x - 2z = 3\,.$$

Solution. Let $g(x,y,z) = x^2 + y^2 - z^2$ and

$$h(x,y,z) = x - 2z - 3\,.$$

The objective function is the square root of

$$f(x,y,z) = x^2 + y^2 + z^2\,,$$

subject to $g(x,y,z) = 0 = h(x,y,z)\,.$

Take $\nabla f = \lambda \nabla g + \varphi \nabla h$, or $\nabla f - \lambda \nabla g = \varphi \nabla h$

we have

$$2x(1-\lambda) = \varphi \quad (1)$$
$$2y(1-\lambda) = 0 \quad (2)$$
$$2z(1+\lambda) = -2\varphi \quad (3)$$
$$x^2 + y^2 - z^2 = 0 \quad (4)$$
$$x - 2z = 3 \quad (5)$$

From (2), we have either $y = 0$ or $\lambda = 1$.

$$(2z+3)^2 + y^2 - z^2 = 0 \Rightarrow z^2 + 12z + y^2 + 9 = 0 \, (6)$$

Case 1. $y = 0$. Then solving (4) and (5):

$$\left.\begin{array}{r} x^2 = z^2 \\ x - 2z = 3 \end{array}\right\} \text{ implies that } (x, y, z) = (-3, 0, -3)\,,\ (1, 0, -1)\,.$$

Case 2. $\lambda = 1$. Then (1) implies that $\varphi = 0$ and so $z = 0$ from (3).

So, (5) gives $x = 3$ and this will then violate (4). So, this case is impossible.

Answer: max distance $\sqrt{f(-3, 0, -3)} = 3\sqrt{2}$ at $(-3, 0, -3)$;

min distance $\sqrt{f(1, 0, -1)} = \sqrt{2}$ at $(1, 0, -1)$.

Alternative. We can solve this question by first merging (4) and (5) into one:

$$(2z+3)^2 + y^2 - z^2 = 0\,.$$

Let $G(y,z) = (2z+3)^2 + y^2 - z^2 = z^2 + 12z + y^2 + 9\,.$

The objective function is the square root of

$F(y,z) = (2z+3)^2 + y^2 + z^2 = 5z^2 + 12z + y^2 + 9\,,$

subject to $G(y,z) = 0\,.$

$$10z = 2\lambda z \Rightarrow z = 0, \lambda = 0$$

$$2y = 2\lambda y \Rightarrow y = 0, \lambda = 0$$

Case 1. $z = 0 = y$, this is obviously a contradiction.

Case 2. $z = 0 = \lambda$, then $(2z+3)^2 + y^2 - z^2 = 0 \Rightarrow 9 + y^2 = 0$, also a contradiction.

Case 3. $y = 0 = \lambda$, then $(2z+3)^2 - z^2 = 0 \Rightarrow 2z + 3 = \pm z \Rightarrow z = -3, -1$. So,

$$(x, y, z) = (-3, 0, -3)\,,\ (1, 0, -1)\,.$$

Case 4. $\lambda = 0$ only. Then $y = z = 0$, which is Case 1, a contradiction.

Example 8

Use Lagrange multipliers to find the maximum and minimum values of the function $f(x,y,z) = x^2 + y^2 - \frac{1}{20}z^2$ on the curve of intersection of the plane $x + 2y + z = 10$ and the paraboloid $x^2 + y^2 = z$.

Solution. Let $g(x,y) = x + 2y + z - 10$ and $h(x,y) = x^2 + y^2 - z$ so that $g(x,y) = 0 = h(x,y)$. Take $\nabla f = \lambda \nabla g + \varphi \nabla h$.

$$\begin{cases} 2x = \lambda + 2x\varphi & (1) \\ 2y = 2\lambda + 2y\varphi & (2) \\ -\frac{1}{10}z = \lambda - \varphi & (3) \end{cases} \qquad \begin{aligned} x + 2y + z &= 10 \quad (4) \\ x^2 + y^2 - z &= 0 \quad (5) \end{aligned}$$

$(1)-(2)\times\frac{1}{2}: 2x-y=(2x-y)\varphi \Rightarrow y=2x$ or $\varphi=1$.

$(4)+(5): x^2+y^2+x+2y=10$ (6).

Case 1. $y=2x$ sub into (6): $5x^2+5x-10=0 \Rightarrow x=1,-2$.

So, we have $(x,y,z)=(1,2,5),(-2,-4,20)$.

Case 2. $\varphi=1$. Then either (1) and (2) imply that $\lambda=0$,

which then implies that (from (3)) $z=10$.

Sub this z into (4) and (5).

Sub this z into (4) and (5).

$$x + 2y = 0$$
$$x^2 + y^2 = 10$$

Solving, we get $y = \pm\sqrt{2}$.

So $(x, y, z) = (-2\sqrt{2}, \sqrt{2}, 10), (2\sqrt{2}, -\sqrt{2}, 10)$.

Point (x, y, z)	$(1, 2, 5)$	$(-2, -4, 20)$	$(-2\sqrt{2}, \sqrt{2}, 10)$	$(2\sqrt{2}, -\sqrt{2}, 10)$
Value of f	15/4	0	5	5
Nature of the point		minimum	maximum	maximum

Case 3. Both $y = 2x$ and $\varphi = 1$. Then similar to Case 2, $z = 10$.

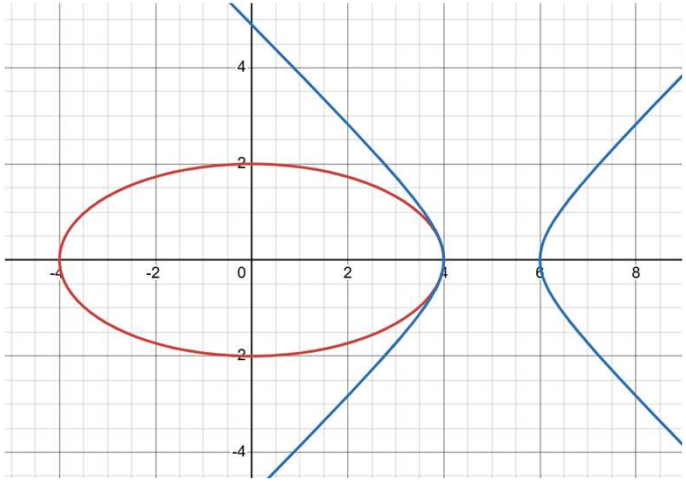
Now if we introduce $z = 10$ and $y = 2x$ (4) and (5),

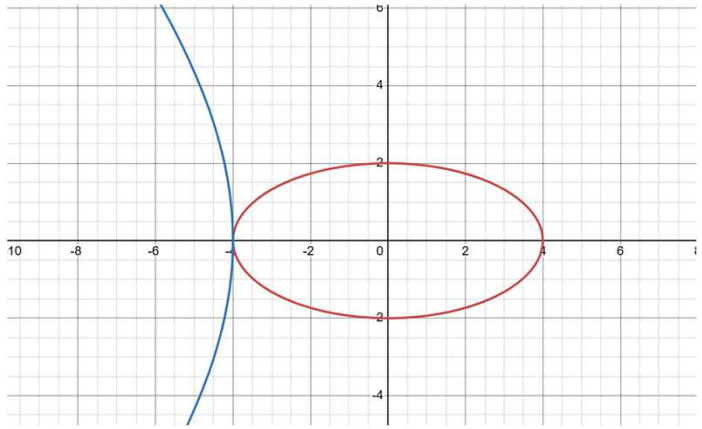
$$\begin{aligned} x + 2(2x) + 10 = 10 &\Rightarrow x = 0 \\ x^2 + 4x^2 - 10 = 0 &\Rightarrow x^2 = 2 \end{aligned}$$

, a contradiction.

Exercise (a)

Find the maximum and minimum of $f(x,y) = x^2 - 10x - y^2$ on the ellipse whose equation is $x^2 + 4y^2 = 16$.



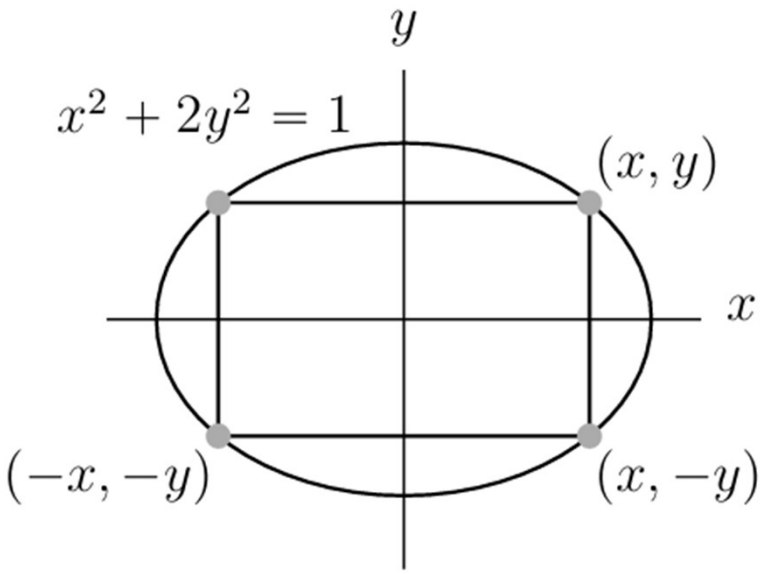


Answer. $f_{\min}(4,0) = -24, \quad f_{\max}(-4,0) = 56$

Exercise (b)

Find the rectangle of the largest area with sides parallel to the coordinates axes that can be inscribed in the ellipse $x^2 + 2y^2 = 1$.

Answer. Maximum area is $\sqrt{2}$.



Exercise (c)

Find the point on the sphere $x^2 + y^2 + z^2 = 1$ that is (i) nearest; (ii) farthest from (1,2,3).

Answer. The nearest point is $\frac{1}{\sqrt{14}}(1,2,3)$.

The farthest point is $-\frac{1}{\sqrt{14}}(1,2,3)$.

Summary

- Apply Lagrange multiplier and you obtain more than one answer, then the biggest one among them is the max and the smallest one is the min.
- The constraint(s) can be substituted into the objective function or other constraint(s).
- Be careful when simplify any equation, if possible do not cross out potential answers (during factorisation).
- For distance question, if you let the objective function without the square root, do remember to add on when you have obtained the solution.
- If you obtain only one solution, you need to further verify it is a minimum/maximum.